12

Magnetic Fields and Electromagnetism

12.1

Magnetic Fields

12.1.1

Properties of a Magnetic Field and Magnetic Field Maps

CHAPTER-OPENING QUESTION—Guess now!

Which of the following can experience a force when placed in the magnetic field of a magnet?

(a) An electric charge at rest.
(b) An electric charge moving.
(c) An electric current in a wire.
(d) Another magnet.

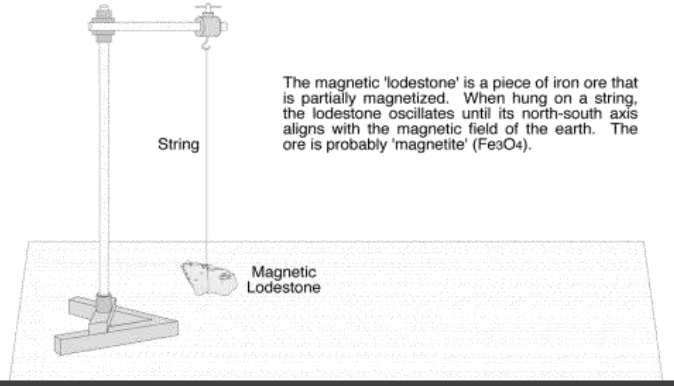
Pliny the Elder

The leading distinction in magnets is the sex, male and female, and the next great difference in them is the colour. Those of Magnesia, bordering on Macedonia, are of a reddish black; those of Breotia are more red than black; and the kind that is found in Troas is black, of the female sex, and consequently destitute of attractive power.

AZ QUOTES

MAGNETIC FIELDS. **D+30+0**

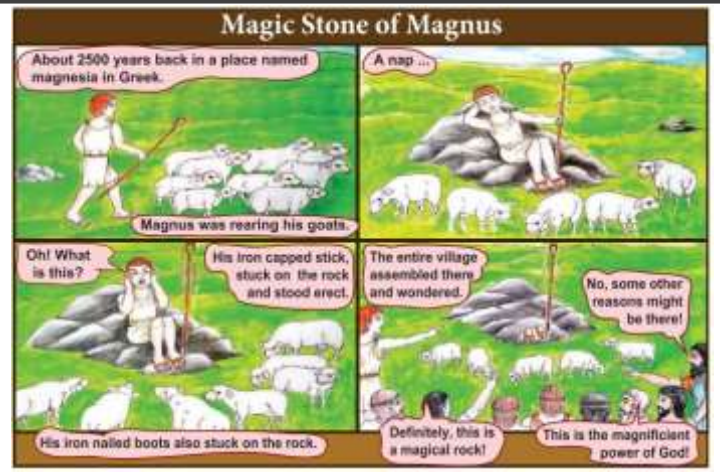
Suspended magnetic lodestone on string.



The magnetic 'lodestone' is a piece of iron ore that is partially magnetized. When hung on a string, the lodestone oscillates until its north-south axis aligns with the magnetic field of the earth. The ore is probably 'magnetite' (Fe_3O_4).

Magnets

- A special stone first discovered < 2,000 years ago in Greece, in a region called "Magnesia," attracted iron, they called it "magnetite" hence the "magnet" name.
- About 1,000 years ago they noticed that a hanging magnet always pointed to the North Star (also known as the "Lodestar". Hence the other name for naturally occurring magnets – "lodestone."



Background

Domains

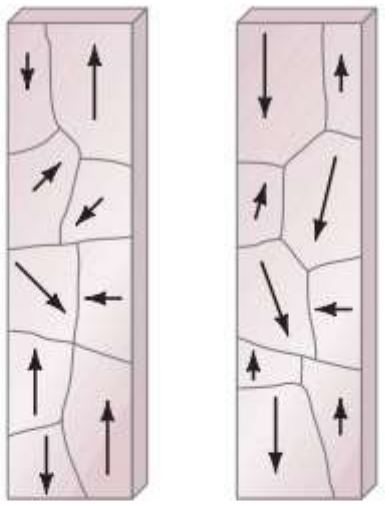


FIGURE 20–42 (a) An unmagnetized piece of iron is made up of domains that are randomly arranged. Each domain is like a tiny magnet; the arrows represent the magnetization direction, with the arrowhead being the N pole. (b) In a magnet, the domains are preferentially aligned in one direction (down in this case), and may be altered in size by the magnetization process.

FIGURE 20–2 Like poles of two magnets repel; unlike poles attract.

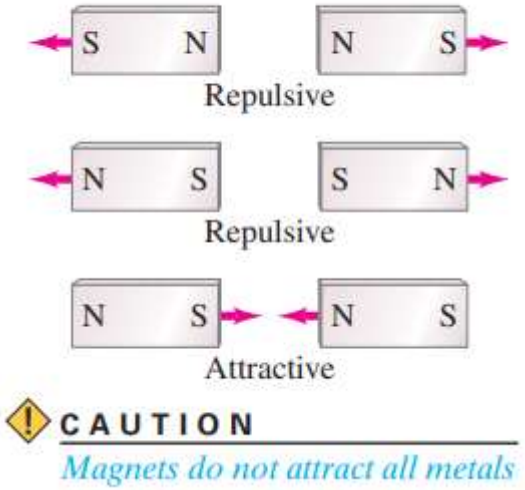
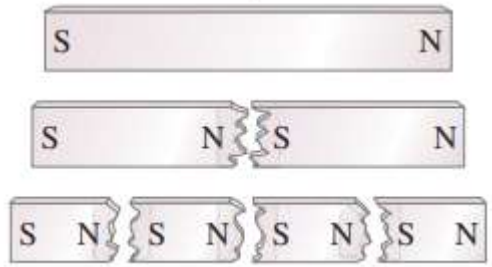
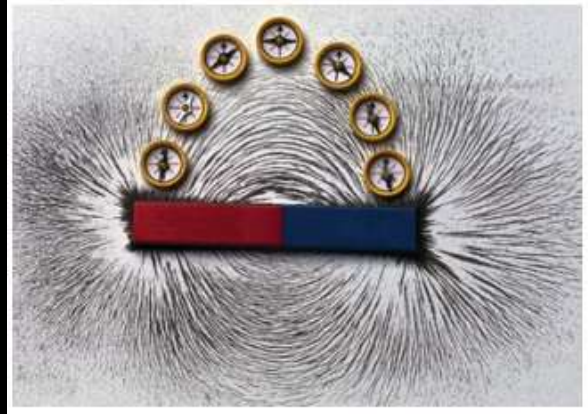
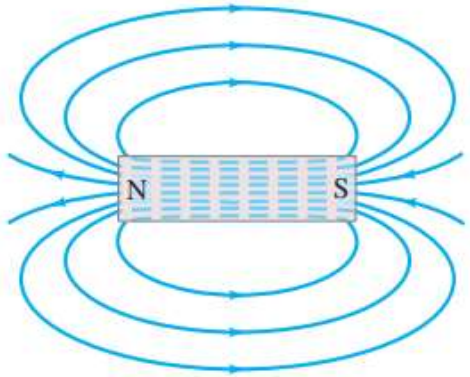


FIGURE 20–3 If you split a magnet, you won't get isolated north and south poles; instead, two new magnets are produced, each with a north and a south pole.





(a)



(b)

FIGURE 20–4 (a) Visualizing magnetic field lines around a bar magnet, using iron filings and compass needles. The red end of the bar magnet is its north pole. The N pole of a nearby compass needle points away from the north pole of the magnet. (b) Diagram of magnetic field lines for a bar magnet.

Definitions:

poles: two ends/faces where magnetic effects are the strongest

North/ South Pole: the two "sides" of a magnet. The field lines point from North to South

Magnetic monopole: a "one-sided" magnet. Does not exist

Ferromagnetic: metals, alloys, oxides with strong magnetic effects

Magnetic Field: the area around magnetic poles where forces occur

Magnetic Field Lines

1. the direction of the magnetic field is tangent to a field line at any point
2. the number of lines per unit area is proportional to the strength of the magnetic field

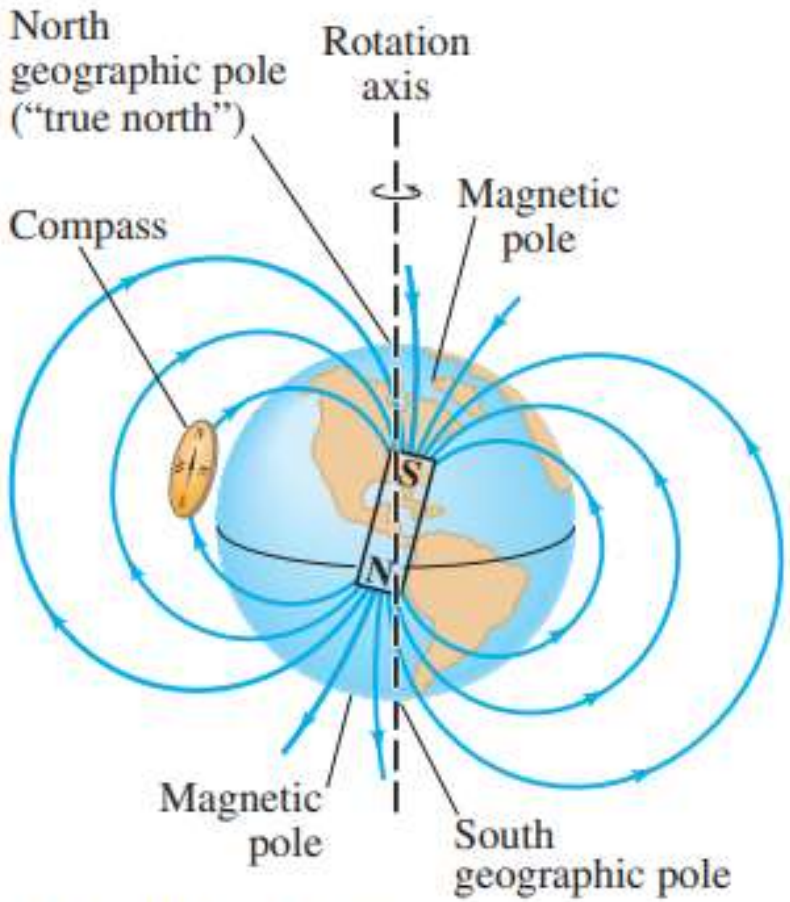
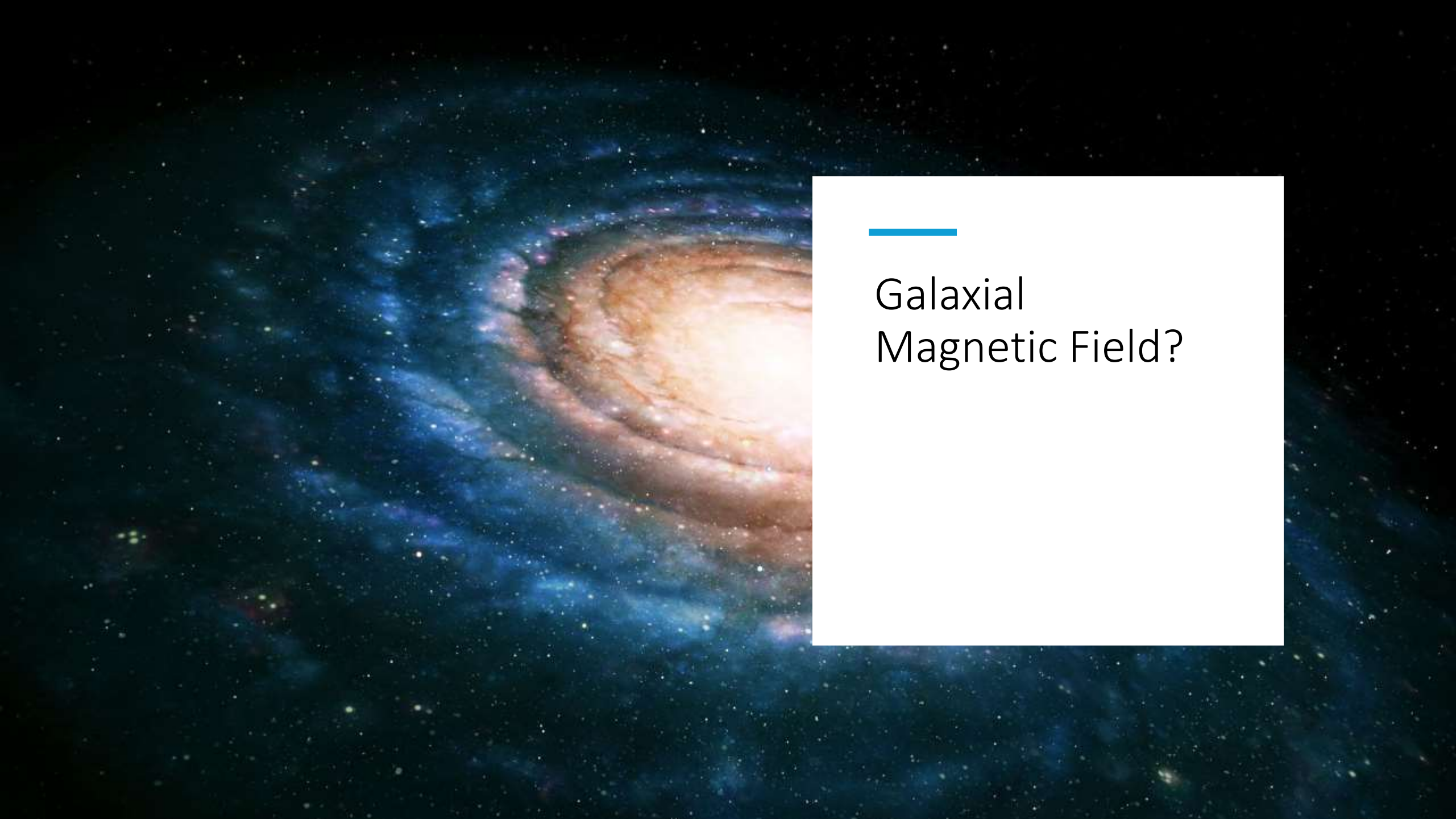


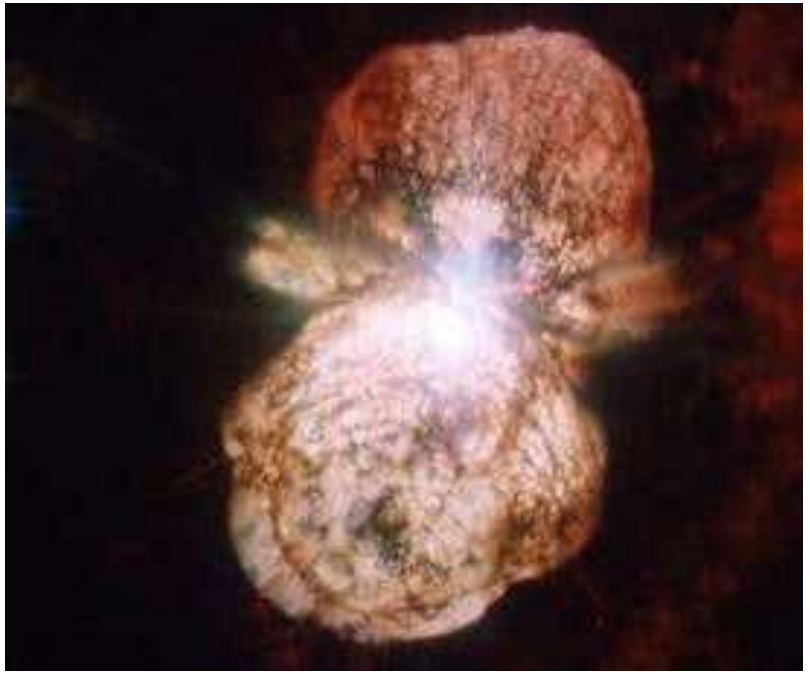
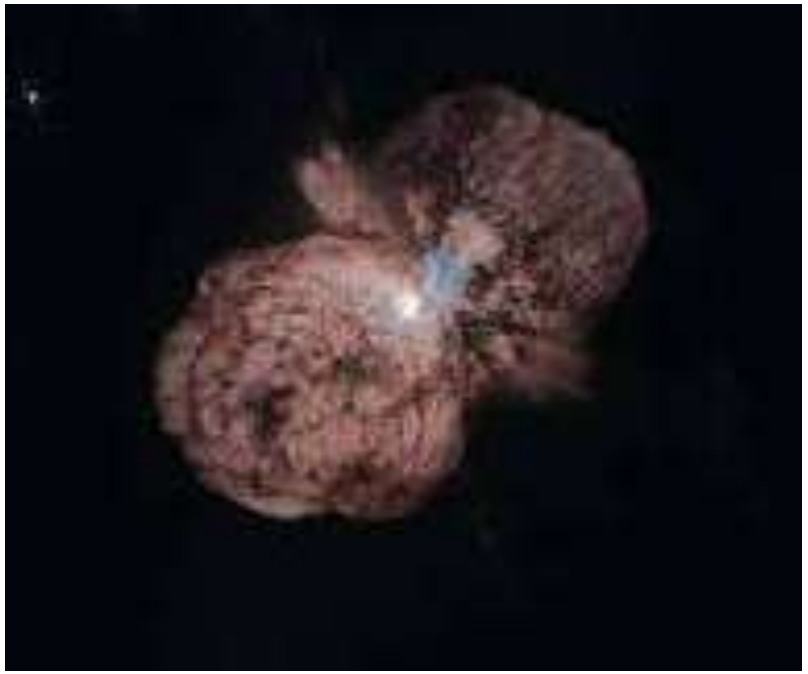
FIGURE 20–5 The Earth acts like a huge magnet. But its magnetic poles are not at the geographic poles (on the Earth's rotation axis).

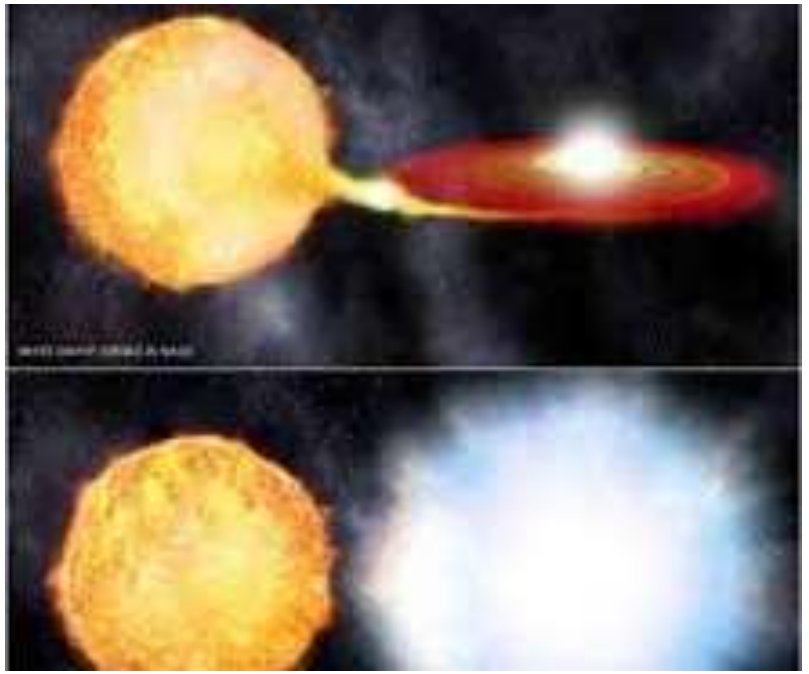
Earth's Magnetic Field:

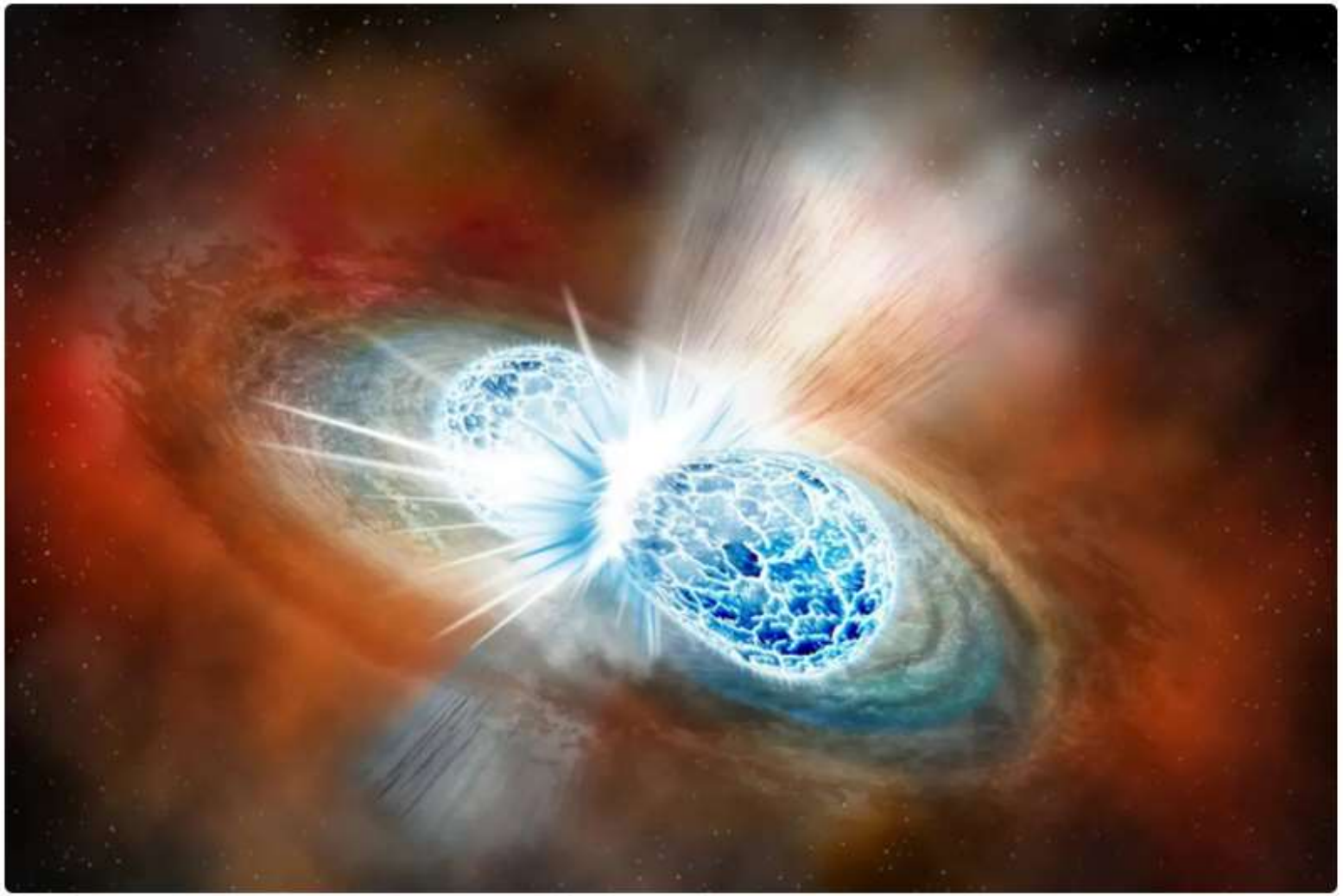
- Magnetic pole: The location that a north magnet will point towards (is actually a <u>South Magnetic Pole)</u>
- Geographic Poles: The locations along the rotational axis
- True North: North Geographic Pole
- Magnetic Declination: the angular difference between "True North" and "North Pole" (0-20°)
- Angle of Dip: The vertical direction the field points at a location on the earth's surface (55° in Miami)



Galaxial Magnetic Field?



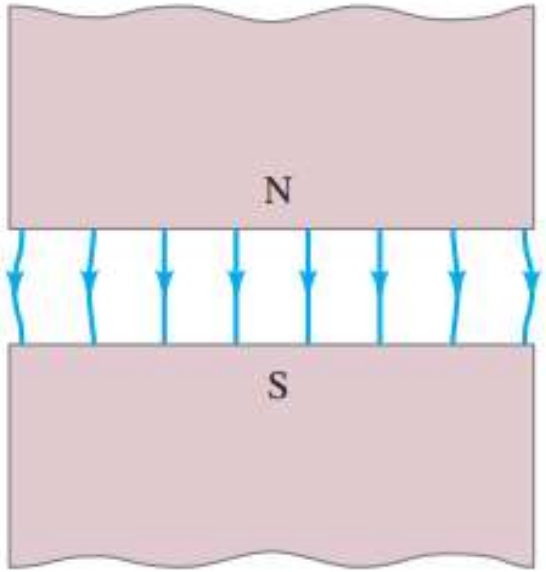


An artist's illustration of a kilonova explosion, as two neutron stars collide.

Robin Dienel/Carnegie Institution for Science)

Uniform Magnetic Field

FIGURE 20–7 Magnetic field between two wide poles of a magnet is nearly uniform, except near the edges.



Uniform Magnetic Field

The simplest magnetic field is one that is uniform—it doesn't change in magnitude or direction from one point to another. A perfectly uniform field over a large area is not easy to produce. But the field between two flat parallel pole pieces of a magnet is nearly uniform if the area of the pole faces is large compared to their separation, as shown in Fig. 20–7. At the edges, the field "fringes" out somewhat: the magnetic field lines are no longer quite parallel and uniform. The parallel evenly spaced field lines in the central region of the gap indicate that the field is uniform at points not too near the edges, much like the electric field between two parallel plates (Fig. 17–1).

[†]Earth's north magnetic pole has been moving over time, on the order of 10 km per year in recent decades. Magnetism in rocks solidified at various times in the past (age determined by radioactive dating—see Section 30–11) suggests that Earth's magnetic poles have not only moved significantly over geologic time, but have also reversed direction 400 times over the last 330 million years. Also note that a compass gives a false reading if you are standing on rock containing magnetized iron ore (as you move around, the compass needle is inconsistent).

Magnetic Fields

- A **magnetic field** is a vector field that describes the magnetic force exerted on moving charges, electric currents, or magnetic materials.
- Magnetic fields are created by **magnetic dipoles** or combinations of dipoles.
- A magnetic dipole has both **north** and **south** polarity.
- Magnetic fields are **not** created by monopoles.

Magnetic Field Lines

- Magnetic field lines allow us to **visualize** the magnetic field around a magnetic dipole or a magnetic object.
- Magnetic field lines point from a north pole to a south pole.
- Magnetic field lines will always form a closed loop through a dipole.

Gauss's Law

Gauss's Law for electric fields states that the electric field through a closed surface is directly proportional to the charge enclosed by that surface. This is Maxwell first equation:

$$\oint \boldsymbol{E} \cdot \boldsymbol{dA} = \frac{\boldsymbol{q_{enclosed}}}{\boldsymbol{\varepsilon_0}}$$

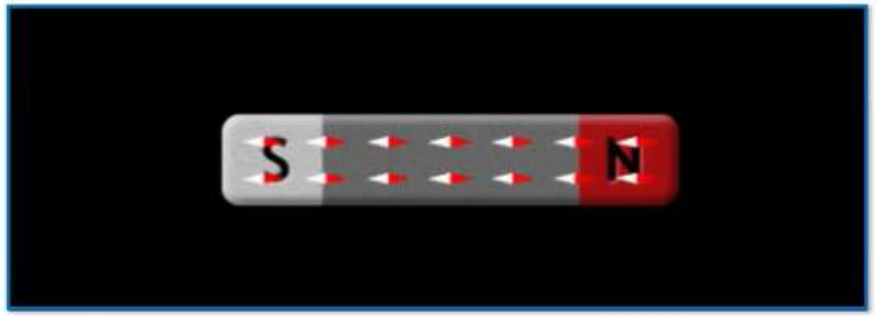
Because magnetic poles always occur as dipoles, the total of any dipole in an enclosed surface will be zero.

This is **Gauss's Law** for magnetism, Maxwell's second equation:

$$\oint \boldsymbol{B} \cdot \boldsymbol{dA} = \boldsymbol{0}$$

Magnets

A bar magnet is composed of multiple magnetic dipoles and has a **north** pole and a **south** pole.



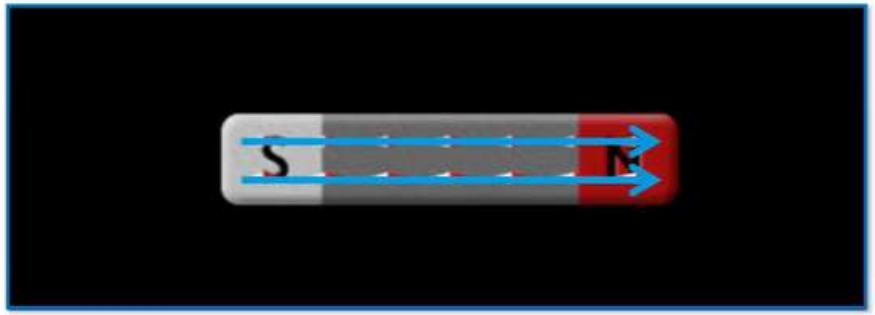
PhET Interactive Simulations University of Colorado Boulder https://phet.colorado.edu

"DOMAINS"!

The magnetic field inside of a bar magnet connects the **north** and **south** poles of the dipoles inside of the magnet.

Magnets

The magnetic field **inside** of a bar magnet points from **south** to **north**.

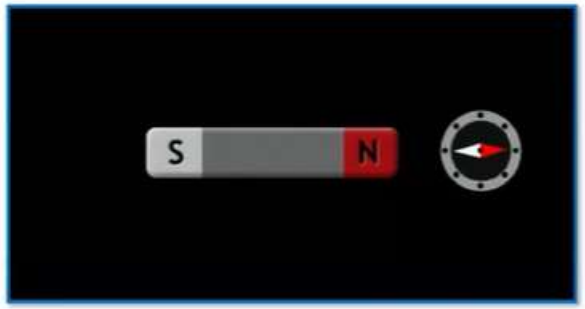


Magnets

A compass will align itself with a **magnetic field** and show the field's direction at a point in space.

Magnets

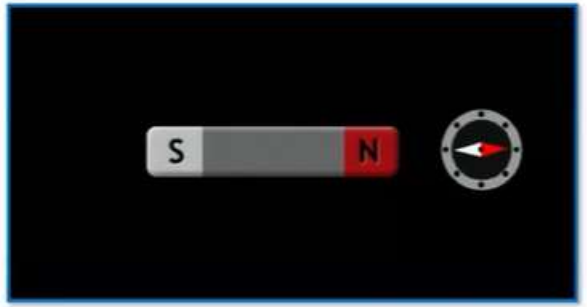
By moving a compass around a magnet, we can see the direction of the magnetic field **outside** of the magnet.



PhET Interactive Simulations University of Colorado Boulder https://phet.colorado.edu

Magnets

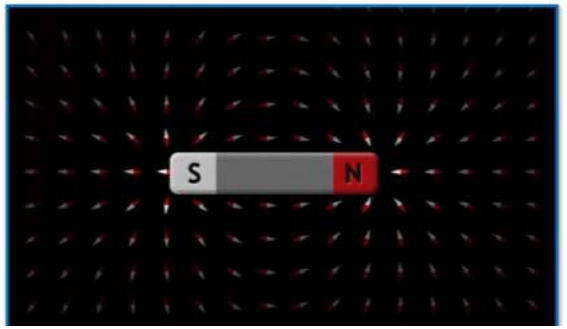
By moving a compass around a magnet, we can see the direction of the magnetic field **outside** of the magnet.



PhET Interactive Simulations University of Colorado Boulder https://phet.colorado.edu

Magnets

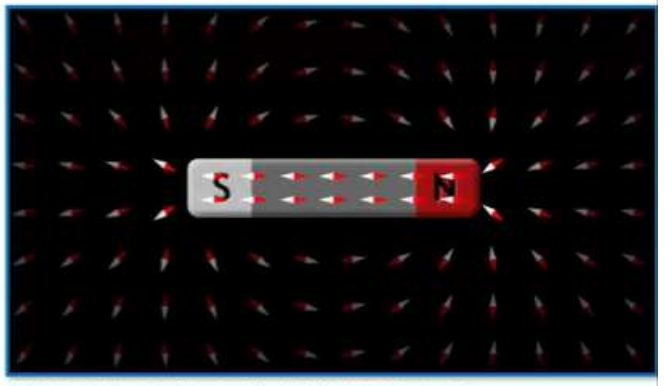
Repeating these measurements over and over will yield a diagram showing the overall shape of the magnetic field.



Inside a magnet, magnetic field lines point from **south** to **north**.

Outside a magnet, magnetic field lines point from **north** to **south**.

Magnetic field lines form closed loops.



PhET Interactive Simulations University of Colorado Boulder https://phet.colorado.edu

Magnetic Field Lines

I have placed a bar magnet below a piece of paper.

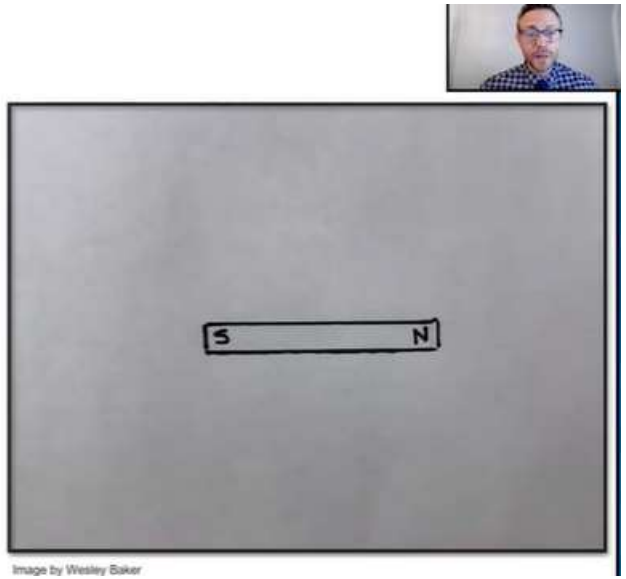
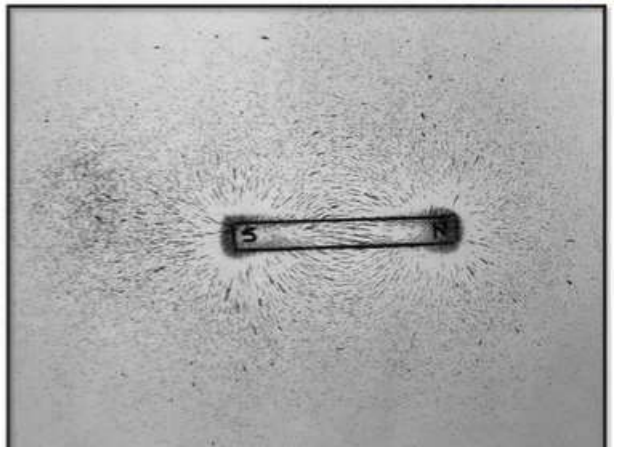


Image by Wesley Baker

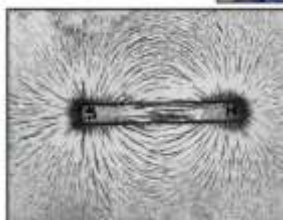
Magnetic Field Lines

Iron filings sprinkled onto the paper show the magnetic field around the bar magnet.

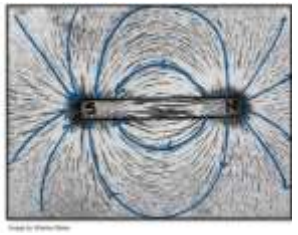


Magnetic Field Lines

After sprinkling a lot of iron filings onto the paper, we can really see the magnetic field.

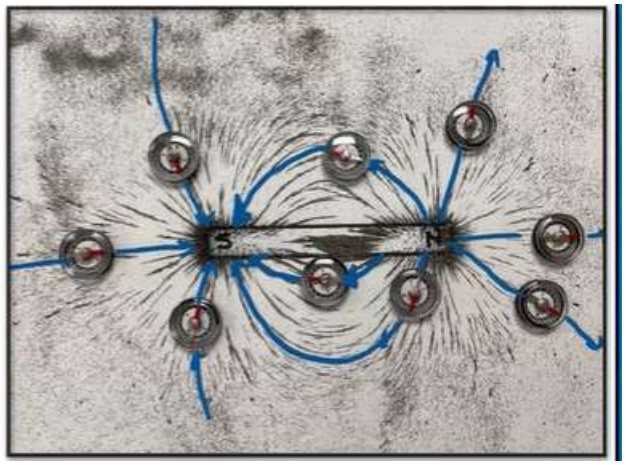


Here is a drawing of the magnetic field lines.



Compasses placed around the magnet align themselves to the magnetic field.

Compasses placed around the magnet align themselves to the magnetic field.



Takeaways

- Magnetic fields are created by magnetic dipoles and describe the direction of the magnetic force at various points in space.
- Magnetic field lines form closed loops.
- Inside a magnet, magnetic field lines point from **south** to **north**.
- Outside a magnet, magnetic field lines point from **north** to **south**.

12.1.2

-Magnetic Field Created by a Magnetic Dipole

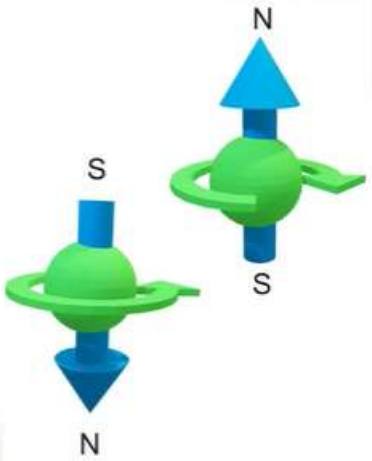
-Effect of a Magnetic Field on Magnetized Objects

Magnetic Dipoles

Magnetic dipoles are created by the rotational motion of electric charge carriers.

Spin is a fundamental property of an electron. Electron spin can be up or down.

The spin of an electron creates a magnetic field, as shown.



Remember: Electrons are negative!

Magnetic Dipole Moments

Magnetic dipoles moments are also created by the circular motion of electric charge carriers.

In this example, an electron moves in a circle. As it does so, it creates a magnetic dipole moment as shown.

Magnetic Dipoles

Because the circular or rotational motion of charge carriers create magnetic dipoles, magnetic poles always occur in pairs.

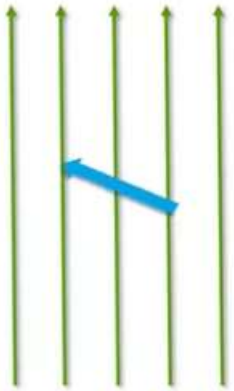
North and **south** poles cannot be separated from each other.

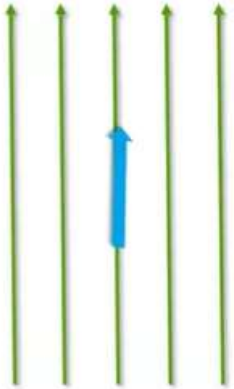
Magnetic Dipoles

A magnetic dipole in an external magnetic field may experience a torque from the magnetic field.

This torque can cause the dipole to align with the magnetic field.

This is similar to the action of a compass in a magnetic field.

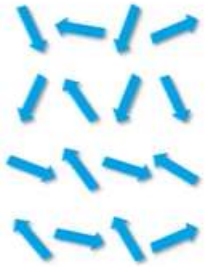




Magnetism in Materials

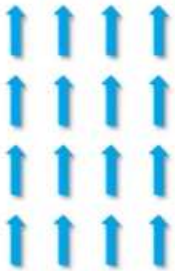
Magnetic dipole moments exist in all materials.

The magnetic moments in nonmagnetic materials point in random directions.



These randomly arranged magnetic dipoles cancel each other out, and these materials do not exhibit a net magnetic field.

The magnetic moments in permanent magnets point in the same direction.



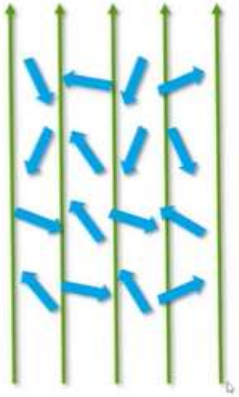
The magnetic moments add together, and these materials exhibit a net magnetic field.

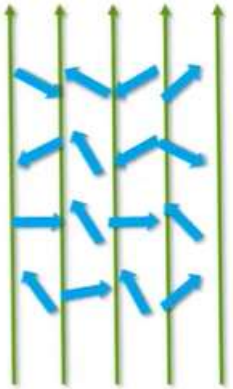
Magnetic Materials

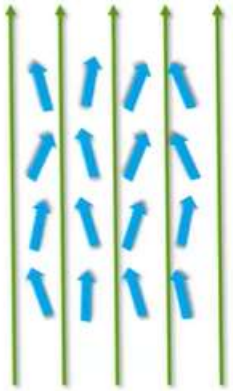
In some materials, magnetic dipole moments can move to align themselves in the presence of an external magnetic field.

These materials exhibit magnetism only the presence of an external magnetic field.

This is called **induced magnetism**.



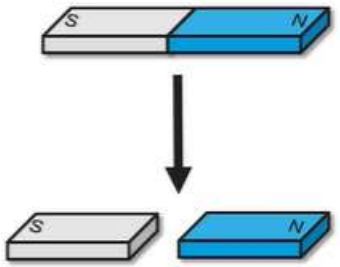




Breaking a Bar Magnet

A group of students are considering breaking a bar magnet in half.

Some of the students think that one half will be a **south** pole and the other half will be a **north** pole, as shown to the right.



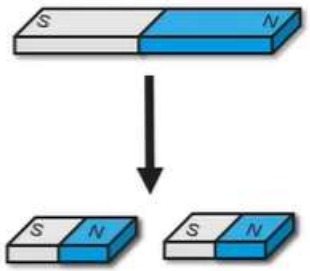
Explain to the students what will really happen to the magnet pieces.

A bar magnet is a permanent magnet composed of aligned magnetic dipole moments.

Each dipole moment has two poles, a north and a south pole. These poles cannot be separated.

When the bar magnet breaks, it will become two magnets, each with a north and a south pole.

This will be true no matter how many times the magnet is broken; each piece will have a north and a south pole.



Earth's Magnetic Field

The magnetic field around the Earth is caused by the motion of charge carriers inside the Earth's core.

What is the current polarity of the "North Pole"?

- The Earth's **geographic** "North Pole" is a **magnetic** south pole.
- The north pole of a compass or any isolated magnet is attracted to the Earth's "North Pole."
- **Opposite poles attract**: north poles are attracted to south poles.
- When a compass aligns itself with an external magnetic field, its north end points in the direction of the field. Magnetic fields point away from north magnetic poles and toward south magnetic poles.

Takeaways

- Magnetic dipoles result from the rotational motion of electric charges.
- **North** and **south** poles are never found in isolation.
- Permanent magnets have dipoles that are always aligned.
- Magnetism can be induced in a material when its magnetic dipoles align with an external magnetic field.
- Like magnetic poles repel, and opposite poles attract.

12.1.3

-The effect a Material’s Composition has on its Magnetic field

-Definition of Magnetic Permeability

Magnetic Permeability

$$\mu$$

Magnetic permeability describes a substance's ability to form internal magnetic fields.

Magnetic Permeability

Empty space has its own magnetic permeability. It is known as the **permeability of free space**:

$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{N}/A^2$$

Note: $N / A^2 = (T\ m) / A$

This is a fundamental physical constant that is important when quantifying the magnetic field created by a moving charge carrier.

The permeability of free space is relatively low, suggesting that free space does not create an internal magnetic field.

Relative Permeability

By looking at the ratio of a material's permeability to the permeability of free space, we can determine that material's **relative permeability**.

$$\mu_r = \frac{\mu}{\mu_0}$$

A material's relative permeability gives us an indication of how magnetic that material is.

Nonmagnetic materials have relative permeabilities very close to 1 and can be classified as **diamagnetic** or **paramagnetic**.

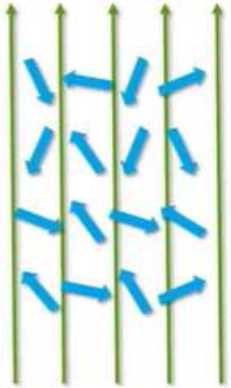
If $\mu_r \approx 1$, $\mu \approx \mu_r$

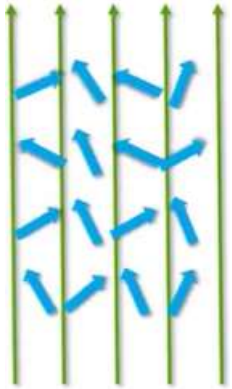
Paramagnetism

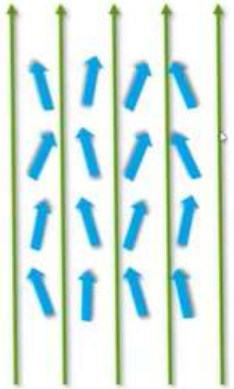
In most materials, the net magnetic dipole moment from an **electron's spin** is cancelled by a paired electron with opposite spin.

In some materials with unpaired or free electrons, these magnetic dipole moments do not cancel. These unpaired electrons have a net magnetic dipole moment.

Paramagnetism





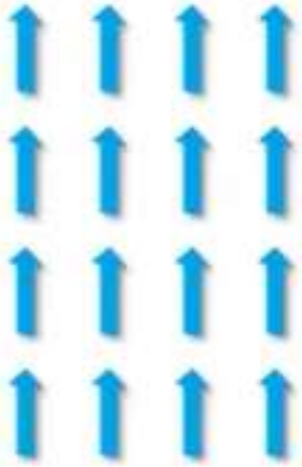


These unpaired magnetic dipole moments will align with an external magnetic field when it is present.

Paramagnetic materials, like aluminum, gold, and copper are magnetic in the presence of a magnetic field and are nonmagnetic in the absence of one.

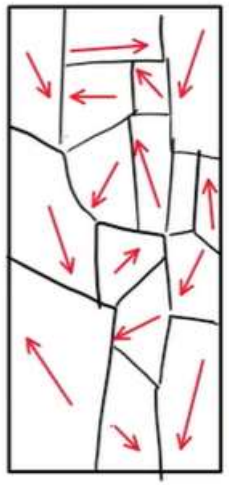
Ferromagnetism

The magnetic moments in a ferromagnetic material align in the absence of external magnetic fields.



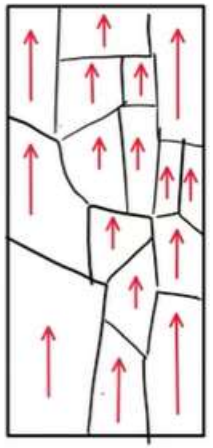
These areas of alignment are called **magnetic domains**.

These domains can have random alignment, canceling each other out and giving the object a net-zero magnetic field.



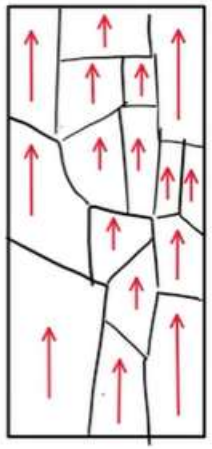
Ferromagnetism

In the presence of an external magnetic field, magnetic domains align with each other.



Ferromagnetism

In the presence of an external magnetic field, magnetic domains align with each other.



The internal magnetic field created by these aligned domains is very strong.

This internal magnetic field increases the external magnetic field.

Strengthening an Electromagnet

Stacked loops of wire connected to the ends of a battery create a weak magnetic field. A much stronger magnetic field is created when the same loops of wire (still connected to the battery) are wrapped around an iron nail.

Is the nail paramagnetic or ferromagnetic? In your justification, discuss the nail's relative permeability.

Strengthening an Electromagnet

The nail is **ferromagnetic**.

The magnetic domains in ferromagnetic materials align in the presence of an external magnetic field, strengthening the external magnetic field.

The iron nail increases the magnetic field created by the loops of wire and the battery.

Iron has a high relative permeability.

Relative permeability describes a material's ability to form internal magnetic fields.

The nail was able to form a **strong** internal magnetic field in the presence of an external magnetic field.

Takeaways

Magnetic permeability describes a substance's ability to form internal magnetic fields.

Paramagnetism is weak attraction to a magnetic field that some materials experience due to magnetic dipole moments created from electrons' spin.

Ferromagnetic substances have an inherent magnetic field due to the alignment of their magnetic domains.